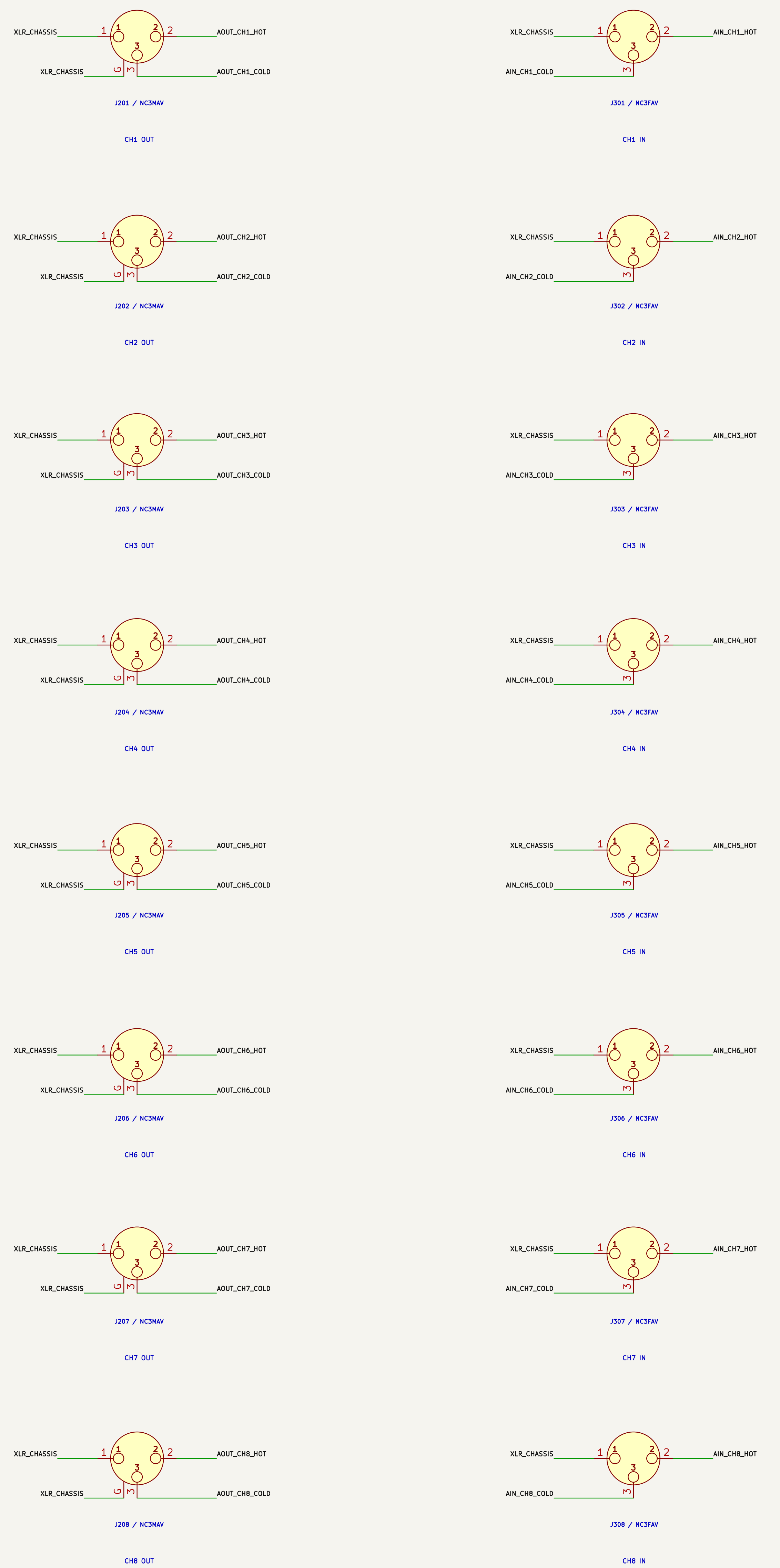
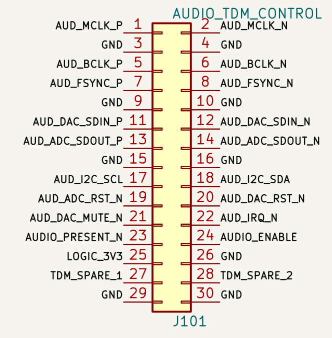
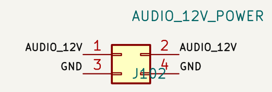
A. PANEL–SUPPORTED BALANCED XLR BANK



B. CARRIER INTERFACES





Local filtered conversion produces +/−15 V, AKM 5 V analog and audio 3.3 V.

Mates with CM5-CARRIER J201; 100 ohm differential pairs with interleaved grounds.

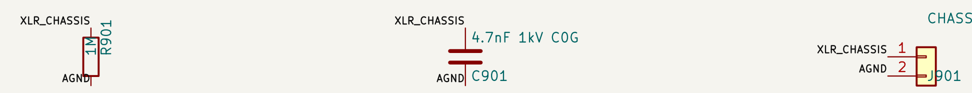
C. INPUT SIGNAL CHAIN

NC3FAV -> chassis/RFI/ESD -> THAT1206 -> level/filter network -> AK5558VN

Eight differential channels; no phantom power; +4 dBu nominal / +24 dBu max.

AK5558VN analog interface values must be recalculated from the current datasheet.

F. XLR SHIELD / CHASSIS BOND



Every XLR pin 1 and NC3MAV shell G bonds directly to the local chassis plane.

R901/C901 are the single controlled RF/static bond to AGND; verify values in EMC test.

No audio return current may use the XLR_CHASSIS copper or panel fasteners.

D. OUTPUT SIGNAL CHAIN

AK4458VN -> reconstruction/gain filter -> OPA165x -> THAT1646 -> RFI/fault network -> NC3MAV

Stable into 600 ohm test load; 10 kohm or higher is the normal operating load.

Hardware mute remains asserted through boot, clock loss, reset and power transfer.

E. CAPTURE SHEETS TO ADD

01 LVDS TDM receivers/transmitter, I2C/control protection and clock options

02 AK5558VN supplies, straps, decoupling and eight ADC input filters

03 AK4458VN supplies, straps, decoupling and eight DAC output filters

04 Eight THAT1206 balanced input receivers and connector protection

05 Eight OPA165x/THAT1646 balanced output stages and fault protection

06 AUDIO_12V conversion, +/−15 V, low-noise AKM rails and sequencing

07 Chassis/pin-1 bonding, shield strategy, star points and test fixtures

NOT A FABRICATION RELEASE – AKM networks require datasheet calculation
Left bank male outputs; right bank female inputs
+4 dBu nominal, +24 dBu maximum, active-balanced XLR stages
AK5558VN ADC and AK4458VN DAC over buffered differential TDM

ProComm

Sheet: /
File: Audio-8x8.kicad_sch

Title: Radxa CM5 ProComm Audio 8x8 – Interface Contract

Size: A2	Date: 2026-08-13	Rev: A0
KiCad E.D.A. 10.0.5-rc1		Id: 1/1